

Hypoglycemic episodes and risk of dementia in older patients with type 2 diabetes mellitus.

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Evidence abstract

CONTEXT Although acute hypoglycemia may be associated with cognitive impairment in children with type 1 diabetes, no studies to date have evaluated whether hypoglycemia is a risk factor for dementia in older patients with type 2 diabetes.

OBJECTIVE To determine if hypoglycemic episodes severe enough to require hospitalization are associated with an increased risk of dementia in a population of older patients with type 2 diabetes followed up for 27 years.

DESIGN, SETTING, AND PATIENTS A longitudinal cohort study from 1980-2007 of 16,667 patients with a mean age of 65 years and type 2 diabetes who are members of an integrated health care delivery system in northern California.

MAIN OUTCOME MEASURE Hypoglycemic events from 1980-2002 were collected and reviewed using hospital discharge and emergency department diagnoses.

Cohort members with no prior diagnoses of dementia, mild cognitive impairment, or general memory complaints as of January 1, 2003, were followed up for a dementia diagnosis through January 15, 2007.

Dementia risk was examined using Cox proportional hazard regression models, adjusted for age, sex, race/ethnicity, education, body mass index, duration of diabetes, 7-year mean glycated hemoglobin, diabetes treatment, duration of insulin use, hyperlipidemia, hypertension, cardiovascular disease, stroke, transient cerebral ischemia, and end-stage renal disease.

RESULTS At least 1 episode of hypoglycemia was diagnosed in 1465 patients (8.8%) and dementia was diagnosed in 1822 patients (11%) during follow-up; 250 patients had both dementia and at least 1 episode of hypoglycemia (16.95%).

Compared with patients with no hypoglycemia, patients with single or multiple episodes had a graded increase in risk with fully adjusted hazard ratios (HRs): for 1 episode (HR, 1.26; 95% confidence interval [CI], 1.10-1.49); 2 episodes (HR, 1.80; 95% CI, 1.37-2.36); and 3 or more episodes (HR, 1.94; 95% CI, 1.42-2.64).

The attributable risk of dementia between individuals with and without a history of hypoglycemia was 2.39% per year (95% CI, 1.72%-3.01%).

Results were not attenuated when medical utilization rates, length of health plan membership, or time since initial diabetes diagnosis were added to the model.

When examining emergency department admissions for hypoglycemia for association with risk of dementia (535 episodes), results were similar (compared with patients with 0 episodes) with fully adjusted HRs: for 1 episode (HR, 1.42; 95% CI, 1.12-1.78) and for 2 or more episodes (HR, 2.36; 95% CI, 1.57-3.55).

CONCLUSIONS Among older patients with type 2 diabetes, a history of severe hypoglycemic episodes was associated with a greater risk of dementia.

Whether minor hypoglycemic episodes increase risk of dementia is unknown.

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Evidence checklist

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2. **OBJECTIVE** To determine if hypoglycemic episodes severe enough to require hospitalization are associated with an increased risk of dementia in a population of older patients with type 2 diabetes followed up for 27 years.
3. **DESIGN, SETTING, AND PATIENTS** A longitudinal cohort study from 1980-2007 of 16,667 patients with a mean age of 65 years and type 2 diabetes who are members of an integrated health care delivery system in northern California.
4. **MAIN OUTCOME MEASURE** Hypoglycemic events from 1980-2002 were collected and reviewed using hospital discharge and emergency department diagnoses.
5. Cohort members with no prior diagnoses of dementia, mild cognitive impairment, or general memory complaints as of January 1, 2003, were followed up for a dementia diagnosis through January 15, 2007.
6. Dementia risk was examined using Cox proportional hazard regression models, adjusted for age, sex, race/ethnicity, education, body mass index, duration of diabetes, 7-year mean glycated hemoglobin, diabetes treatment, duration of insulin use, hyperlipidemia, hypertension, cardiovascular disease, stroke, transient cerebral ischemia, and end-stage renal disease.
7. **RESULTS** At least 1 episode of hypoglycemia was diagnosed in 1465 patients (8.8%) and dementia was diagnosed in 1822 patients (11%) during follow-up; 250 patients had both dementia and at least 1 episode of hypoglycemia (16.95%).
8. Compared with patients with no hypoglycemia, patients with single or multiple episodes had a graded increase in risk with fully adjusted hazard ratios (HRs): for 1 episode (HR, 1.26; 95% confidence interval [CI], 1.10-1.49); 2 episodes (HR, 1.80; 95% CI, 1.37-2.36); and 3 or more episodes (HR, 1.94; 95% CI, 1.42-2.64).
9. The attributable risk of dementia between individuals with and without a history of hypoglycemia was 2.39% per year (95% CI, 1.72%-3.01%).
10. Results were not attenuated when medical utilization rates, length of health plan membership, or time since initial diabetes diagnosis were added to the model.
11. When examining emergency department admissions for hypoglycemia for association with risk of dementia (535 episodes), results were similar (compared with patients with 0 episodes) with fully adjusted HRs: for 1 episode (HR, 1.42; 95% CI, 1.12-1.78) and for 2 or more episodes (HR, 2.36; 95% CI, 1.57-3.55).
12. **CONCLUSIONS** Among older patients with type 2 diabetes, a history of severe hypoglycemic episodes was associated with a greater risk of dementia.
13. Whether minor hypoglycemic episodes increase risk of dementia is unknown.

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